# Jaeyong Ko

M.S. Student, Industrial Engineering, Seoul National University
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## Research Interests

Visual Reasoning, Unified Multimodal Model, LLM Reasoning, LLM Evaluation

## Education

**Seoul National University**

- **M.S.** *Industrial Engineering* (Advisor: Prof. Pilsung Kang) Sep. 2025 – Aug. 2027 (Expected)
  GPA: 3.74/4.0
- **B.S.** *Industrial Engineering* Mar. 2019 – Aug. 2025
  GPA: 3.72/4.0
  Thesis: Multi-modal Financial Time-series Forecasting by Reprogramming Large Language Models

## Publications

*: Equal contribution

**Preprint**

**[1] Cliff Tokens: Analyzing Failure Trigger Tokens in LLM Mathematical Reasoning**
**Jaeyong Ko**, Jinu Lee, Pilsung Kang, Yukyung Lee.
*Preprint [project page] [paper] [github] [dataset]*

**Publications**

**[2] Oddin: Latent Visual Reasoning in Unified Multimodal Models via Self-Distillation from Image Generation**
**Jaeyong Ko***, Hankyeol Kim*, Pilsung Kang.
*LVR @ WACV 2027 Workshop. [paper]*

**[1] FinLifeBench: Exhaustive Life-Event History and Financial-State Reconstruction from Longitudinal Banking Dialogue**
Hangyeul Lee*, Juyoung Oh*, **Jaeyong Ko**, Sunmin Kim, Jaeik Park, Hyunkyu Kim, Jungmin Son, Pilsung Kang.
*ACM ICAIF 2026. [paper]*

**Special Tracks**

**[2] Can a model think VISUALLY?**
**Jaeyong Ko**, Eunseo Kim, Doehyeon Lee.
*NeurIPS 2026 Education Track.*

**[1] AI Safety Lives in a Direction: Reading, Breaking, and Repairing Refusal in Multimodal Models**
Doehyeon Lee, Yoo-Min Jung, **Jaeyong Ko**.
*NeurIPS 2026 Education Track.*

## Research Experience

**Research Intern**

- **École Polytechnique de Montréal** (Advisor: Prof. Walid Jomaa) Apr. 2024 – Aug. 2024
  Human-centric Smart Manufacturing Workcells for Industry 5.0 applications.
- **KAIST** (Advisor: Prof. Woo Chang Kim) Dec. 2023 – Feb. 2024
  Reinforcement learning applications for goal-based wealth management.

## PROJECTS

**Development of Core Foundation Model and AI Agent Technology for Thermal-Process Manufacturing** | *Granted by Ministry of Trade, Industry and Energy* Sep. 2026 – Present

- Designed the backbone architecture for the thermal-process foundation model and established an integrated learning framework utilizing domain data and PINN.

**Duo 2.0: Evaluating LLM Reliability in Long Financial Conversations** Dec. 2025 – Oct. 2026

| *Granted by KakaoBank*

- Quantified LLM capabilities in resolving ambiguous user queries using LLM-as-a-Judge methodologies.
- Designed a financial life-cycle event conversation benchmark for domain-specific conversational reliability.

**Development of Advanced Integrated Monitoring Technologies for Laser Welding** Jul. 2025 – Dec. 2025

| *Granted by LG Electronics*

- Designed and conducted experiments for a laser welding anomaly detection model utilizing time-series and image multimodal data.

**SNU Startup Consulting Group (ACT)** *President & Team Member* Mar. 2023 – Dec. 2023

- Led a student consulting group and secured four startup consulting engagements with *Moderntec*, *Designovel*, *Spring Onward*, and *Sweetspot*.
- Developed GTM strategies for *Designovel* (AI fashion-tech), *Moderntec* (EV charging infrastructure), and *Wavebridge* (cryptocurrency OTC platform).

## INDUSTRY EXPERIENCE

**Product Manager Intern** *EXGRAM* Jul. 2022 – Oct. 2022

Planned app configurations and push notifications for the app, Veloo.

## TEACHING EXPERIENCE

**Peer Tutor**

- Production Control, Seoul National University (406.310) Sep. 2024 – Dec. 2024
- Human Factors Engineering, Seoul National University (406.304) Sep. 2023 – Dec. 2023

## HONORS & AWARDS

**Master's Student Research Grant (1 year)** | *National Research Foundation of Korea* 2026

- Project: Cliff Tokens: Identifying Single-Token Failure Triggers in LLM Mathematical Reasoning

**Outstanding Paper Award** | Poster Presentation, 2026 KDMS Summer Conference 2026

**15th Lee Junhan Award (USD 2,200)** | Undergraduate Academic Category 2023

**3rd Place, Startup Idea Competition** | *College of Engineering and College of Business, SNU* 2019

**3rd Place, IE Idea Competition** | *Korean Institute of Industrial Engineers* 2019

## SKILLS

- **Programming and Software:** Python, R, MySQL, Docker, Git, LaTeX.
- **Languages:** Native Korean; professional working proficiency in English (TOEFL 5.0/6.0, OPIc IH).

## MILITARY SERVICE

**Corporal** *Republic of Korea Army* Jan. 2021 – Jul. 2022

Completed full-term mandatory military service.

*(Updated Oct 2026)*